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“Deep Learning for Photomultiplier Crystal Detection Signals”

Miguel Escudero Rodríguez

David Izquierdo-García
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Deep Learning for Photomultiplier Crystal Detection Signals

Miguel Escudero Rodríguez
Master's Degree in Machine Learning for Health
Universidad Carlos III de Madrid, Leganés, Spain
Email: 100569281@alumnos.uc3m.es

***Abstract*—Spatial resolution is a general limitation of Positron Emission Tomography (PET), and monolithic scintillator crystals are one of the routes towards improving it: they preserve the continuous distribution of the scintillation light instead of segmenting it, so the interaction position can be estimated between sensors rather than at a fixed pitch. That advantage carries a dependency: the light of every event is shared across many channels of the silicon photomultiplier (SiPM) array, so a single faulty channel distorts the reconstructed position of events across the whole crystal, and current practice is to detect the fault and discard either the affected events or the module. We propose recovering the lost signal instead. HEXGNN is a graph neural network of 38 305 parameters that takes the hexagonal array as its structure, with one node per sensor and one edge between adjacent sensors, and estimates the charge of a suppressed channel from the sixty that remain, using the geometry of the detector as a prior instead of learning it from the data. It needs no faulty hardware to train: we suppress a channel of a healthy module and take its recorded value as the label. We used flood acquisitions with a ^{22}Na source on 159 physical modules of a hexagonal PET detector, measured rather than simulated, and evaluated on modules the network had never seen. Imputation removes 56.6 ± 2.3 % of the positioning error introduced by the fault, leaving a residual displacement of 0.28 mm at P90, and preserves the energy spectrum to 84.7 ± 0.9 %. Training with one to four simultaneous failures extends the result to several dead channels at no cost in the single-channel case. The correction is applied before the centre of gravity is computed, so it returns a complete 61-channel vector that the rest of the processing chain consumes unchanged.**

***Index Terms*—Positron emission tomography, silicon photomultiplier, missing data imputation, deep learning, graph neural networks,silicon photomultiplier failure.**

1 INTRODUCTION

Positron Emission Tomography (PET) is a non-invasive molecular imaging modality and one of the central techniques of nuclear medicine, providing an *in vivo* assessment of the metabolic and biochemical state of a patient [1]. Unlike purely anatomical modalities, PET relies on radiotracers: biologically active molecules chemically labelled with positron-emitting radionuclides [2]. Once administered, the tracer distributes according to the physiological pathway targeted by its carrier molecule, so that the measured signal reflects function rather than structure. When an emitted positron annihilates with an electron in the surrounding tissue, two photons of 511 keV — the rest mass energy of the electron — are released in nearly opposite directions. Detecting both of them in coincidence defines a line of response (LOR) along which the annihilation took place, and the accumulation of millions of LORs allows the three-dimensional distribution of activity to be reconstructed [1]. This detection scheme gives PET a functional sensitivity that few other non-invasive imaging techniques can match [3], at the price of the tracer itself: positron-emitting radionuclides are radioactive by nature, costly to produce, and short-lived, which ties any PET facility to a nearby cyclotron and specific facilities [2].

This functional readout is most valuable where disease alters metabolism before it alters anatomy. In oncology, ^{18}F-FDG PET has become a cornerstone for diagnosis, staging, restaging and the assessment of treatment response [4], [5]. Its role in neurology is arguably even more distinctive, since it reveals pathologies that produce no macroscopic structural abnormality on conventional Magnetic Resonance Imaging (MRI) or Computed Tomography (CT), such as the characteristic hypometabolic and amyloid patterns of Alzheimer's disease [6], among other neurodegenerative and cerebrovascular disorders.

The sensitivity of PET is not matched by its spatial resolution. Clinical scanners typically resolve 4–5 mm full width at half maximum (FWHM) [8], and recent advances push the number towards 3 mm [8], [9], higher than the numbers normally achieved with MRI or CT. This limit decomposes into four contributions, two physical and two belonging to the detector, which behave very differently from one another [10]. The physical ones are the finite distance the positron travels before annihilating, set by the emitting isotope [11], and the small angular deviation from antiparallel emission,

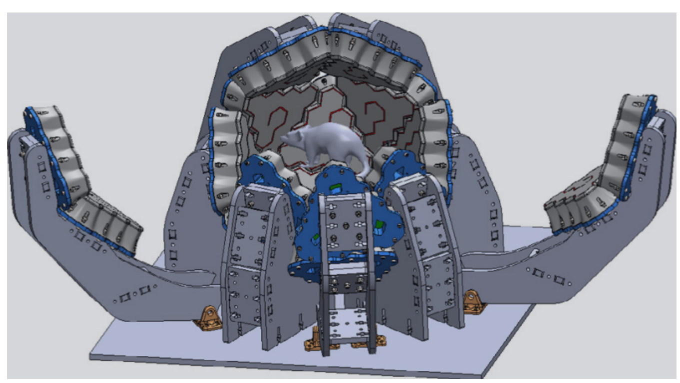

Figure 1: Three-dimensional model of the quasi-spherical, hexagonal-based PET scanner. Reproduced from [7].

caused by the residual momentum of the electron–positron pair. Neither can be engineered away, although the second translates into a positional error that scales with the diameter of the detector ring and is therefore smaller in compact systems. The detector terms are more tractable. In pixelated arrays, the size of the individual element fixes the finest scale at which an interaction can be localised; shrinking it improves resolution, but the inter-crystal reflectors then occupy a growing fraction of the detector volume, and the probability that a photon deposits energy across several elements rises. The decoding error — the uncertainty in determining where within the module the interaction occurred — amounts to roughly one third of the detector width [10]. Unlike positron range and non-collinearity, this term is not a physical floor but a consequence of how the light distribution is read and interpreted, and it is therefore the one most open to improvement, whether through detector geometry, readout electronics or the algorithms that interpret the signal. This is the ground occupied by organ-dedicated scanners. Adapting the detector geometry to a specific anatomical region increases solid-angle coverage, and with it sensitivity and resolution, at a lower manufacturing cost than a general-purpose whole-body system; designs exist for breast, brain, prostate and cardiac imaging [12]. The trade-off is that irregular geometries complicate calibration and data correction, and that small gantry apertures accentuate parallax errors and degrade the uniformity of spatial resolution across the field of view [12]. Much of the design effort in these systems therefore shifts onto the detector itself.

Our group is currently developing a quasi-spherical brain-dedicated PET scanner along these lines [7]. Rather than the cylindrical ring of conventional systems, its detector modules tile a closed surface around the head (Fig. 1), approaching full 4π steradian coverage and thereby collecting a far larger fraction of the emitted coincidences [7]. The modules are hexagonal, a shape chosen for two independent reasons: hexagons tile a surface without gaps and approximate a sphere more closely than square modules of comparable size, and hexagonal scintillators have also been shown to offer slightly better light yield and energy resolution than square, triangular or cylindrical prisms [13]. Each module is built around a monolithic LYSO crystal [14].

The choice between segmenting the scintillator and leaving it continuous is not incidental, and it conditions how the interaction position can be recovered at all [15]. In pixelated arrays, the crystal is cut into individual elements separated by reflectors, so that the interaction is assigned to whichever element fired. Event positioning is then simple and fast, but the resolution cannot be finer than the element itself [10], and two costs follow from making the elements smaller: the reflectors occupy a growing fraction of the detector volume, reducing the packing fraction, and the number of readout channels rises. Multiplexing schemes, in which several crystals share a readout line and the position is recovered by decoding the light distribution, are commonly used to keep this cost manageable [12]. Pixelated arrays also provide no intrinsic depth-of-interaction (DOI) information, which has motivated layered designs such as phoswich detectors, where stacked scintillators with different decay times allow a coarse depth assignment at the price of a degraded timing resolution [12].

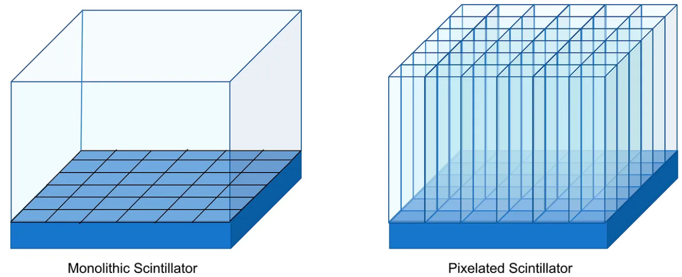

Figure 2: Monolithic (left) versus pixelated (right) scintillator configurations. In the monolithic design a continuous slab is coupled to a segmented photosensor array; in the pixelated design the crystal is cut into elements coupled one-to-one to the photosensors. Reproduced from [17] under CC BY 4.0.

Monolithic scintillators take the opposite approach. A single continuous block is coupled to an array of photodetectors, and the light from each interaction spreads over many channels at once [15]. Positioning is no longer a matter of identifying an element but of estimating a coordinate from the recorded light distribution, which shifts the resolution limit from the geometry of the crystal to the quality of the positioning algorithm, and makes the DOI recoverable as a continuous variable [16]. The absence of internal reflectors also restores the packing fraction lost in segmented designs [12]. This is the configuration adopted in the scanner described above, and it is the light sharing across the array that the present work exploits.

This light distribution is read out by an array of silicon photomultipliers (SiPMs) optically coupled to the rear face of the crystal [15]. Each interaction produces signal in a subset of channels, and a readout application-specific integrated circuit (ASIC) amplifies, thresholds and digitises each of them, so that an event is fully described by the vector of charges recorded across the array. Because a channel only produces an output once it exceeds the threshold set in the electronics, most entries of that vector are zero for any given event and only a few channels contribute; the sparsity of the signal is therefore a property of the acquisition system rather than a defect of the data [18].

Estimating the interaction position from that vector admits solutions of widely varying complexity. The most common is the centre of gravity (CoG), which places the event at the average of the photodetector coordinates, each weighted by the charge that channel recorded [19]; in practice the square of the charge is normally used as the weight, which reinforces the contribution of the channels closest to the interaction point and improves resolution [20]. Its appeal is that it is deterministic, has no parameters to tune and requires no prior calibration. In exchange, it exploits only part of the information contained in the light distribution: recovering the DOI, or correcting the bias the method itself introduces near the crystal edges, calls

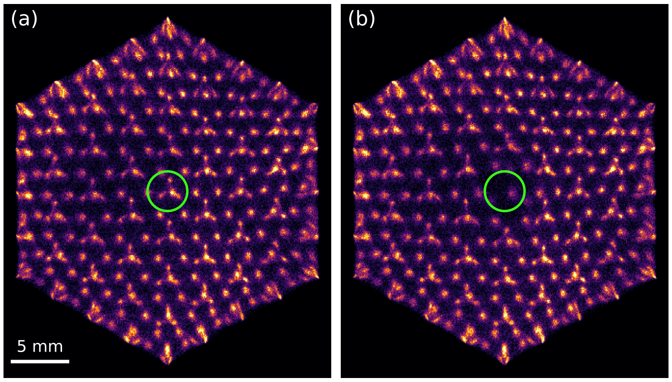

Figure 3: Flood map of the same detector module reconstructed by centre of gravity, using the same events in both panels: (a) with all 61 channels operational and (b) with one channel disabled, marked in green. The structures engraved in the crystal appear as discrete spots; the inset magnifies the neighbourhood of the disabled channel.

for substantially more elaborate algorithms [21].

The two-dimensional histogram of the positions computed in this way over a large number of events forms the flood map (Fig. 3), in which the structures engraved in the crystal appear resolved as discrete spots. It is the image conventionally used to characterise a detector of this kind, and its sharpness depends on all 61 channels being operational: the CoG averages over the whole array, and a single missing term is enough to displace the centroid. The effect is confined to the neighbourhood of the failing channel — in the module of Fig. 3 the reconstructed position shifts by a median of 0.41 mm for events within 4 mm of it, and by less than 0.01 mm elsewhere — but within that neighbourhood the engraved structures are no longer resolved.

Silicon photomultipliers degrade and fail for a variety of reasons: thermal drift, manufacturing defects, deterioration of the optical coupling, or faults in the readout channel. The consequence for the image is well recognised in industrial practice: a dead pixel introduces reconstruction artefacts and incorrect activity values, and commercial systems include quality-control procedures to identify it [22]. The adopted general solution, however, is to discard the affected information and compensate for it downstream of positioning, either by duplicating events from neighbouring pixels in list mode or by interpolating in sinogram space [22]. The channel is not recovered: it is masked out.

This class of problem has motivated the application of deep learning to PET detector readout [23]. The most developed line is the estimation of the interaction position in monolithic crystals, where networks trained on calibration data consistently outperform analytical algorithms and recover the DOI as a continuous variable [24], [25]. Two previous studies address precisely this task on the present detector [18], [26]. All of them share the same framing: the light distribution recorded by the array is taken as input and the network returns a coordinate.

Closer to what is proposed here, deep learning has also been applied to recovering channel signals rather than positions. Networks have been trained to invert multiplexed readout schemes, reconstructing the individual channel values from a reduced set of measurements and validating the result on the recovered flood map [27], [28], and the approach has recently been extended to semi-monolithic detectors, where the full 64-channel response is reconstructed from 16 row-and-column sums by exploiting the light sharing between slabs [29]. In all these cases, however, the information carried by each channel is still present in the acquired data: multiplexing encodes it in a known linear combination, and the network learns to invert a mapping fixed by design. A failed channel is a different situation, since nothing of its signal reaches the acquisition and the only remaining trace of it is the statistical correlation that light sharing induces among its neighbours.

That situation is not unique to PET. Recovering the response of defective elements in a sensor array is an established problem in imaging, addressed either by interpolating from neighbouring pixels or, more recently, by training autoencoders to reconstruct corrupted regions from their surroundings [30]. The formulation adopted here belongs to that family: it is the masked autoencoder setting [31], in which part of the input is deliberately hidden and the model is trained to reconstruct it, which allows supervision to be generated from healthy modules without requiring labelled faulty detectors. What distinguishes the present case is that the redundancy being exploited is physical rather than statistical: the light emitted in a single interaction is measured simultaneously by several photodetectors, so the value of a channel is constrained by the response of those around it.

This work proposes to recover that channel rather than discard it. If the light from each interaction spreads over several photodetectors, the charge that the failed sensor would have recorded is not entirely lost: it remains implicit in what its neighbours measured, and the problem is to estimate it. The difficulty lies less in the model than in the supervision. In a faulty module there is no way of checking whether an estimate is correct, since the true charge of the dead channel is unobservable by definition: if it could be measured, the channel would not be faulty. What cannot be observed, however, can be simulated. It suffices to take a healthy module, disable one of its channels and keep the original value as a reference; the fault is fabricated, and with it the label. The whole approach rests on that idea, and it is applied to experimental data acquired with the detector described above rather than to simulations, so that the model learns the behaviour of the system as it actually is, with its noise, its threshold and its variation between modules.

The choice of model follows from the geometry of the array. The immediate option would be to treat the vector of sixty-one charges as a list of numbers and let a dense network discover on its own which channels are related to which, but that means learning from scratch something that is already known: which sensor is adjacent to which. The photodetectors form a hexagonal tiling in which every interior sensor has six immediate neighbours, and that neighbourhood is precisely the path along which the light is shared. A structure of this kind is naturally described as a graph, with one node per sensor and an edge between each pair of adjacent sensors, and it therefore admits a neural network that operates on it by propagating information between neighbours [32]. The model proposed here follows that approach, incorporating the geometry of the detector as prior knowledge instead of leaving it implicit in the data.

The evaluation is framed along the same lines. The error in the estimated charge is the quantity the model minimises, but on its own it says little about the usefulness of the method: what matters is how much is recovered of what the fault degrades. Two physical effects are therefore measured: the displacement of the reconstructed position, which is the direct consequence on the flood map, and the conservation of the energy spectrum, which determines whether the restored signal is physically admissible. All of this on modules the network has not seen during training, and compared against classical interpolation methods in order to establish that the complexity of the model is warranted. The behaviour of the method when more than one channel fails is also examined, with particular attention to contiguous groups, which are what is observed in real faulty detectors and the case that displaces the reconstructed position the most, since when an entire region disappears the lost sensors are left without healthy immediate neighbours to draw on.

2 Materials and Methods

2.1 Detector and Data

Every measurement reported here comes from detector modules of the scanner described in Section 1. Each module is built around a monolithic LYSO hexagonal prism, 17.6 mm on a side and 13 mm thick, laser-etched into 364 optical structures that constrain how the scintillation light spreads before reaching the photosensor [14]. An array of silicon photomultipliers is optically coupled to its posterior face [7]. The array carries 61 photosensors, one per readout channel, which tile a hexagonal lattice with a pitch of 3.75 mm and span 26 mm across, so that the sensors do not reach the crystal boundary. The lateral faces of the crystal carry an absorbing black adhesive that suppresses edge reflections. This has a consequence that recurs throughout our results: interactions near the perimeter lose a measurable fraction of their light, which shifts both the charge recorded by the peripheral channels and the position of the energy photopeak in those regions.

We used acquisitions recorded with a ^{22}Na source under flood irradiation, one file per physical module, comprising 159 healthy modules and 62 modules with at least one suspected faulty channel. Each file holds on the order of 10^6 events. An event activates 17 channels on average, ranging from a single channel to 54, since a channel only produces an output once it exceeds the threshold set in the readout electronics. We applied neither energy calibration nor an energy window. Both would require a per-channel calibration that is not available for this set of modules, and the metric we use to assess spectral fidelity is constructed so that their absence cancels, as Section 2.6 explains.

We split the healthy modules by file rather than by event, with a fixed seed: 149 for training, 5 for validation and 5 held out for testing. The distinction matters more than it might appear. Each file is a different physical detector, with its own gain, its own optical coupling and its own defects, so splitting by file measures generalisation to hardware the model has never seen. Splitting by event would let the model learn the idiosyncrasies of a detector and then be tested on that same detector, which is a considerably easier problem and not the one that arises in practice. The test partition was reserved before any training and used neither for fitting nor for model selection.

2.2 Preprocessing and Sample Generation

The raw quantity is a vector $\mathbf{r} \in \mathbb{R}^{61}$ holding the digitised charge of each channel for one event. We applied two operations before it reaches the network. Occasional negative values, produced by baseline subtraction in the electronics, were clipped to zero, since a negative charge has no physical meaning and would distort the centroid. Events in which no channel exceeded the threshold carry no information and were discarded.

We then simulated a faulty channel by forcing a subset $\mathcal{D}$ of the entries to zero and asking the model to recover the original vector from what remained. The network additionally receives a binary mask indicating which entries were suppressed, so that a channel switched off is distinguishable from a channel that legitimately recorded no signal. This is the masked autoencoder setting [31] applied to detector channels instead of image patches, and it makes the supervision self-generated: the target is the original vector, so no module with a genuinely dead sensor is needed for training. We drew a fresh $\mathcal{D}$ for every sample, so a given event is never seen twice with the same failure pattern.

Normalisation, however, deserves a comment, because it constrains the architecture. We scaled each event by the maximum of its own masked vector,

$$\mathbf{x} = \frac{\tilde{\mathbf{r}}}{\max_c \tilde{r}_c}, \qquad \mathbf{y} = \frac{\mathbf{r}}{\max_c \tilde{r}_c}, \tag{1}$$

where $\tilde{\mathbf{r}}$ is the masked vector and $\mathbf{y}$ the target. Two choices are embedded here. Scaling per event rather than per dataset follows from the physics: for position estimation what matters is how the light distributes across the array, not how much light the interaction released. Computing the factor *after* masking means that the target of a suppressed channel exceeds unity whenever that channel was the brightest of the event. The output layer is therefore linear, with no sigmoid or clamp, because bounding it would make the brightest failures unrecoverable by construction.

One further choice concerns which channel to suppress. Roughly three quarters of the channels in any given event are below threshold, so drawing the seed uniformly would produce a training set in which most samples require no correction at

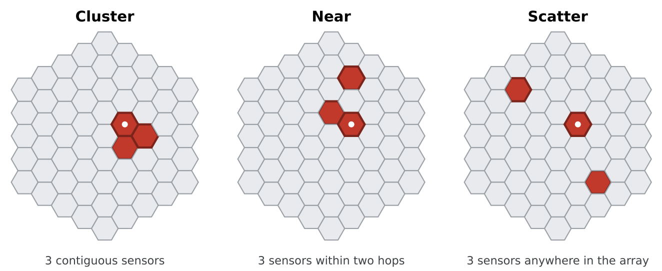

Figure 4: The three failure regimes used to generate the training masks, shown on the 61-sensor array for $k = 3$. Suppressed sensors are marked in red and the white dot indicates the seed from which the pattern grows. Cluster forces the failures to be contiguous, scatter distributes them across the array, and near places them within two hops of the seed without requiring contact.

all, and the model would reasonably learn to predict zero. We therefore balanced the draw: half of the samples suppress a channel that carried signal and half suppress a channel already at zero.

2.3 Failure Regimes

A single dead channel is the simplest fault, but not the only one. When we inspected the flood maps of the 62 defective modules, we found three recurring patterns. What matters is not so much how many sensors fail as how they are arranged: one whose neighbours are all healthy can be reconstructed from them, whereas one surrounded by other dead sensors cannot.

We reproduced those patterns with three mask-generation strategies, illustrated in Fig. 4. In the *cluster* regime the k suppressed channels grow outward from the seed so that they remain contiguous, so an entire region of the array disappears at once and the affected sensors are left without healthy immediate neighbours. In the *scatter* regime they are drawn uniformly across the array, so each failure is effectively isolated. The *near* regime sits between the two: the k channels lie within two hops of the seed without being required to touch. That intermediate case is the one we observed most often in real faulty modules, typically two or three sensors separated by one or two positions, and it is the reason we introduced it rather than settling for the two extremes.

2.4 Network Architecture

The geometry of the array suggests how to model the problem. A dense network receiving the 61 charges as a flat vector would have to learn which sensors are adjacent, something already known from the physical layout. We therefore represented the detector as a graph whose nodes are the 61 active sensors and whose edges join physically adjacent ones, and propagated information along those edges [32], [33]. The resulting adjacency has 312 directed edges, with 37 nodes of degree six, 18 of degree four and 6 of degree three, and it is fully symmetric.

Each layer of our model, which we refer to as HexGNN, updates a node from its own state and the mean of its neighbours,

$$h_c^{(l+1)} = \sigma\left(W_s h_c^{(l)} + W_n \frac{1}{|\mathcal{N}(c)|} \sum_{j \in \mathcal{N}(c)} h_j^{(l)} + b\right), \quad (2)$$

where $\mathcal{N}(c)$ are the neighbours of sensor c and σ is a Gaussian error linear unit (GELU) [34]. The aggregation is isotropic: every neighbour of a node shares the weight matrix W_n, so the layer encodes adjacency but not direction. Batch normalisation [35] is applied between blocks. Our reference configuration uses a hidden width of 48 and four blocks, for 38 305 trainable parameters.

We compared this model against three groups of alternatives. As classical baselines without learned parameters we used the mean of the neighbouring channels and a per-channel linear regression on the remaining 60, both fitted on training modules only and both excluding the target channel from their inputs. As neural baselines without a geometric prior we trained a deep multilayer perceptron and a residual multilayer perceptron with channel attention. Both receive the same information but no notion of adjacency, which isolates the contribution of the prior from that of raw capacity.

The aggregation in Eq. 2 is deliberately simple, and it is fair to ask whether that simplicity is what limits performance. We therefore tested four alternatives, each addressing a different suspicion about where the bottleneck might be.

If the six neighbours of a sensor contribute unequally — and they might, since the light reaching one of them depends on the direction of the interaction — then averaging them discards useful information. Graph attention networks (GAT) [36] learn a weight per neighbour instead of fixing it, which lets the model decide how much each one should count. A related but broader concern is that the mean is only one summary of a neighbourhood. Principal neighbourhood aggregation (PNA) combines several statistics and scales them by node degree, retaining information that any single one throws away [37]; degree varies here between three and six, so the scaling is not vacuous.

The other two alternatives question the locality of the model rather than its aggregation. Our layers propagate information one hop at a time, so with four blocks a sensor only sees four rings away. If the missing charge were constrained by the global shape of the light distribution, a wider receptive field would help. We tested this in two forms: a graph U-Net, which pools the graph into a coarser representation before expanding it back [38], and a graph transformer (Graphormer) in which every node attends to every other one regardless of adjacency [39]. We also tested metric edge features, anisotropic hexagonal convolution with one weight per direction, per-channel equalisation, heteroscedastic prediction of mean and variance, and averaging ensembles [40].

Table 1: Training configuration of the reference model.

Parameter	Value
Architecture	HexGNN, 4 blocks, hidden width 48
Trainable parameters	38 305
Activation	GELU
Loss	MSE over the 61 channels
Optimiser	AdamW [42]
Learning rate	10^{-3}, cosine decay [43]
Weight decay	10^{-4}
Batch size	512
Training samples	15 943 000 (single pass)
Validation events	400 000, fixed masks
Model selection	Validation MAE on the imputed channel

2.5 Training

We trained with the mean squared error (MSE) computed over the full 61-channel output rather than the suppressed entries alone,

$$\mathrm{Loss}_{\mathrm{MSE}} = \frac{1}{61}\sum_{c=1}^{61}\left(\hat{y}_c - y_c\right)^2, \tag{3}$$

which treats the task as ordinary regression and keeps the healthy channels as a consistency constraint.

We chose it after comparing it against the mean absolute error (MAE) and the Huber loss [41], which interpolates between the two by behaving quadratically for small residuals and linearly for large ones,

$$\mathrm{Loss}_\delta(e) = \begin{cases} \frac{1}{2}e^2 & \text{if } |e| \le \delta, \\ \delta\left(|e| - \frac{1}{2}\delta\right) & \text{otherwise,} \end{cases} \tag{4}$$

with $e = \hat{y}_c - y_c$ and $\delta = 0.1$. The motivation for testing it is that the targets contain legitimate extreme values, since a suppressed channel that was the brightest of its event has a target above unity, and a quadratic penalty weights those cases heavily. In practice the MSE gave the lower bias, which is the property we were unwilling to trade, so it is the loss used throughout.

We also tested two physics-informed variants, one penalising the displacement of the reconstructed position and one penalising the shift of the energy spectrum; both are reported in Section 3.7.

Table 1 lists the configuration. Two aspects of the procedure are worth making explicit. First, the training set does not fit in memory, so modules are loaded one at a time and contribute 107 000 events each; the model therefore performs a single pass over 15 943 000 masked samples. Since the masks are regenerated for every sample, no exact input is ever repeated, which is why we applied no early stopping. Second, we selected the final weights by the validation error on the imputed channel, computed on a fixed set of 400 000 events with a fixed mask seed. We did not use the total loss for this purpose. It is dominated by the 60 healthy channels and stays nearly flat across the run, so it separates checkpoints poorly.

Each run stores its preprocessing configuration inside the checkpoint. This is a safeguard rather than a detail: a model trained with one normalisation and evaluated with another produces meaningless numbers without raising any error, and verifying the match automatically removes that failure mode.

2.6 Evaluation Metrics

The quantity the model minimises is the error in the recovered charge, but on its own it says little about whether the method is useful. What matters is how much of the degradation caused by the fault is undone. We therefore evaluated on three levels: the charge itself, the position it produces, and the physical quantities the detector is built to deliver.

At the charge level we report, over the N events in which channel c was suppressed, the MAE and the bias

$$\mathrm{MAE}_c = \frac{1}{N}\sum_{i=1}^{N}|\hat{y}_i - y_i|, \qquad \mathrm{bias}_c = \frac{1}{N}\sum_{i=1}^{N}(\hat{y}_i - y_i), \tag{5}$$

both in the normalised units of Eq. 1. The bias acts as a guardrail. A model can attain a low MAE while predicting systematically low, which would displace every reconstructed position in the same direction; only the signed average reveals it. Because a normalised error is hard to interpret on its own, we also express the MAE relative to the mean charge that the channel collects. That ratio is sensitive to the number of events used for the denominator, so we computed numerator and denominator on the same sample throughout.

The position level is our primary metric, since it is the one that reaches the image. For each event we computed the interaction position by CoG weighted by the squared charge [19], [20], in three versions of the same event: intact, with the channel suppressed, and with the channel imputed. Comparing them against the intact position gives two displacements. The *degraded* displacement ΔR_{deg} is how far the centroid moves when the fault goes uncorrected, and therefore measures the damage. The *imputed* displacement ΔR_{imp} is how far it still lies from the true position after the model has acted, and therefore measures the residual error. Recovery is the fraction of the damage the model removes,

$$\mathrm{Recovery} = 100 \times \left(1 - \frac{\Delta R_{\mathrm{imp}}}{\Delta R_{\mathrm{deg}}}\right), \tag{6}$$

with both terms evaluated at the same quantile and over the same population of events. A value of 100 % would mean the imputed position coincides with the intact one, and 0 % that imputing changed nothing. We report the value at the 90th percentile (P90), which weights the difficult events, and also the mean. Averaging over the 61 channels gives the macro figures used throughout, and the minimum over channels characterises the worst case. Since a ratio says nothing about the magnitude of what remains, we also report ΔR_{imp} in millimetres.

Two further analyses ask a different question. Recovery tells us how close the imputed position is to the true one, but not whether the restored signal is physically admissible. A model could place events correctly on average while distorting the quantities the detector is built to measure.

The first analysis characterises the *spread* of the positioning error rather than its typical size. Recovery reports a percentile of ΔR; here we summarise the same underlying distribution by its width, taking the error components Δx and Δy and computing a Gaussian-equivalent full width at half maximum from a robust scale estimate,

$$\mathrm{FWHM} = 2.3548 \cdot \frac{1}{2} \left(\hat{\sigma}_{\Delta x} + \hat{\sigma}_{\Delta y} \right), \qquad \hat{\sigma}_v = 1.4826 \cdot \mathrm{MAD}(v), \tag{7}$$

where MAD denotes the median absolute deviation, used so that a few extreme events do not drive the estimate. Alongside it we report the bias $\|(\overline{\Delta x}, \overline{\Delta y})\|$, which separates a systematic displacement of the map from a symmetric blurring around the correct position.

Reporting the width alongside the percentile serves a specific purpose: it splits the error into two components that behave differently. The bias is a systematic shift of the whole distribution and is in principle correctable, because it is predictable from the surviving channels. The width is random scatter around that shift, and no amount of information from the neighbours removes it. A model can be good at one and poor at the other, so we report them separately.

One clarification is needed because the name invites it. This FWHM is not the spatial resolution of the detector: it is computed on the error that the fault introduces, not on the point-spread function of the system, and the two differ by about two orders of magnitude. It is also not independent of Eq. 6, since both summarise the same distribution of positioning errors.

The second concerns energy. Suppressing a channel removes part of the collected light, so the reconstructed energy of the event falls and the photopeak shifts towards lower values; a good imputation should put it back. Measuring this is complicated by the absence of energy calibration, since we cannot convert ADC counts into keV. We avoided the problem with a paired differential design: over the same events we subtract the true charge of the channel and add back the prediction, so any calibration factor multiplies both terms and cancels. We evaluated this by zones of the crystal, defined as concentric hexagonal bands, because the absorbing adhesive on the lateral faces makes the periphery behave differently from the centre [21] and an average over the whole crystal would hide that difference.

Every model is evaluated on the same modules, the same events and the same masks, so that differences between models are measured on paired samples. Section 3.4 quantifies how far that precision extends.

2.7 Implementation Details

We implemented the models in PyTorch, and the data pipeline, the evaluation and the figures in Python with NumPy, SciPy and Matplotlib. Training ran on a single NVIDIA GeForce GTX 1660 SUPER with an AMD Ryzen 5 3600X personal computer.

3 Results

3.1 Imputation Accuracy

This section reports a single suppressed channel, which is the failure mode the method targets; the multi-channel case follows in Section 3.5. Every value is the mean over four independently trained models evaluated on the five held-out modules, and the stated deviation is the spread between those models.

The reference model recovers the charge of a suppressed channel with a MAE of 0.6228 ± 0.0064 in normalised units. Expressed relative to the mean charge that each channel collects, this amounts to a 29.1 % error, and it is not uniform across the array: 25.9 % for channels in the core of the crystal against 34.0 % at the edge. The absorbing adhesive on the lateral faces accounts for that gap, since peripheral channels collect less light and their neighbours carry correspondingly less information about them.

The bias stays at $-0.062{\pm}0.023$, that is, below a tenth of the MAE. This matters because the failure mode we were guarding against is a model that minimises its error by predicting conservatively low values, which would look acceptable in the MAE and would systematically displace the reconstructed position but does not happen in the experiments.

3.2 Recovery of the Reconstructed Position

Table 2: Displacement of the reconstructed position with respect to the intact event, before and after imputation. Mean $\pm$ standard deviation over four independently trained models. The degraded column does not depend on the model and is therefore identical in the four.

	Degraded (mm)	**Imputed (mm)**	**Recovery (%)**
Median	0.099	0.047 ± 0.001	50.22 ± 1.82
Mean	0.241	0.117 ± 0.001	52.15 ± 0.31
P90	0.666	0.283 ± 0.004	58.04 ± 0.37

Charge error is an good indicator, but what reaches the image is the displacement of the centroid, and Table 2 gives it in millimetres. Suppressing a channel moves the reconstructed position of the affected events by a median of 0.099 mm and, for the worst tenth of them, by 0.666 mm. After imputation those figures fall to 0.047 mm and 0.283 mm, so the residual displacement at P90 is 7.5 % of the 3.75 mm sensor pitch. The degraded column is identical across the four models, since it does not depend on the imputation; it serves as a consistency check on the evaluation.

Fig. 5 shows what this means for the image. Each panel is the difference between a flood map and the intact reference, averaged over failures of the sensors along one line of the array. Without correction the difference covers the neighbourhood of the failing channel with a pattern of excess and deficit counts, which is the signature of events being displaced towards the surviving sensors, as the CoG express. After imputation that pattern largely disappears. The lower row, which follows a line towards the edge of the crystal, shows a residual that the upper row does not: the periphery is where the method has least information to work with, as explained in the per-channel errors of Section 3.1.

Performance is not uniform across the detector. Fig. 6 maps the recovery of each of the 61 channels and reproduces the

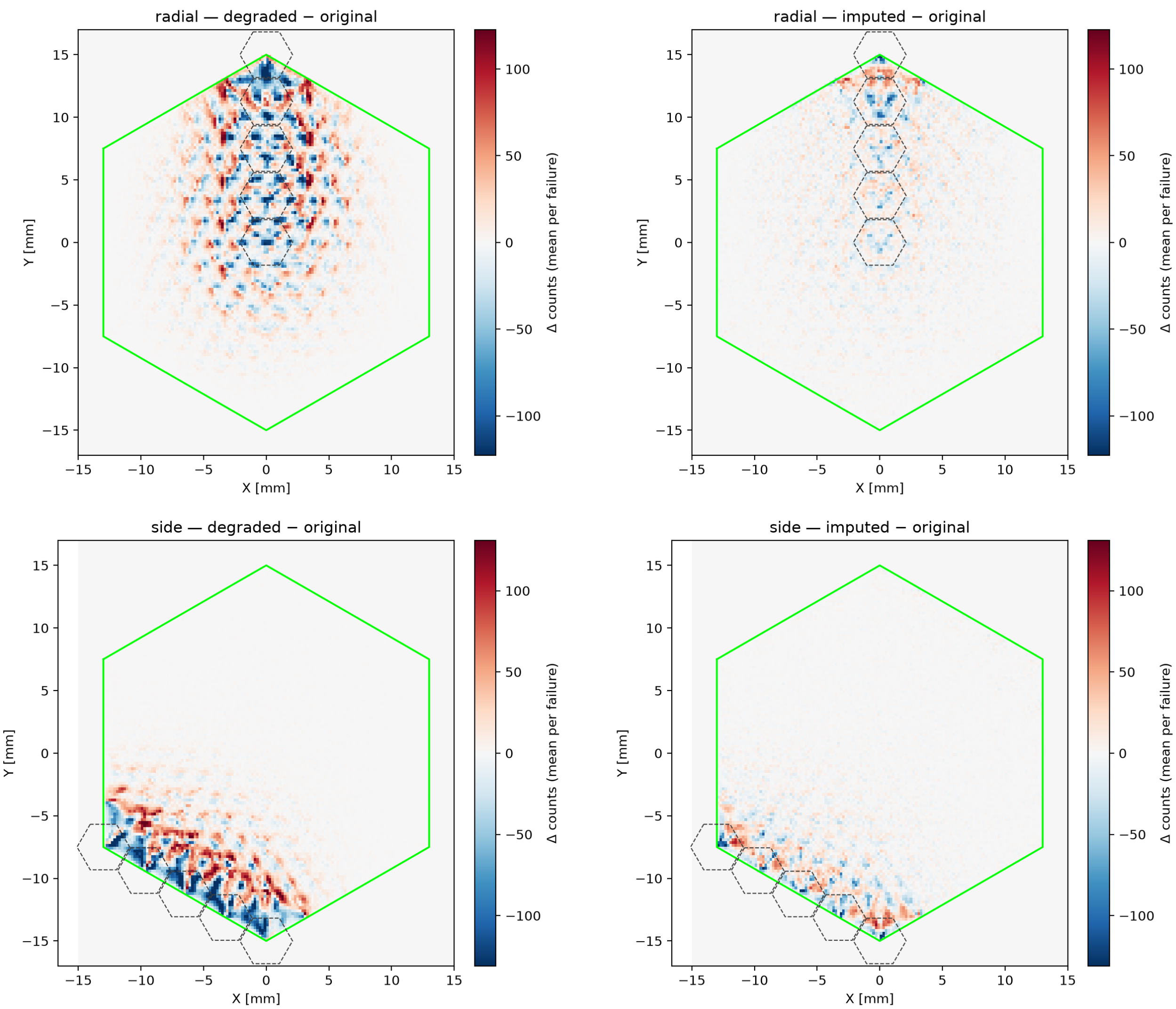

Figure 5: Average flood-map distortion caused by the failure of a sensor. Each panel shows the difference in counts with respect to the intact map: red where the fault adds events and blue where it removes them. Top row, sensors along a radial line; bottom row, sensors along a line towards the edge. Left, without correction; right, after imputation. The dashed hexagons mark the sensors involved and the green outline the crystal boundary.

same core-to-edge gradient. Central channels, whose light is shared with six neighbours, exceed 60 %; channels on the perimeter, with three or four neighbours and less collected light, fall below 45 %.

Fig. 7 closes the sequence with the flood map itself, where the practical consequence is visible: the etched structures around the failing channel, which the fault blurs into one another, are resolved again after imputation.

3.3 *Value of the Geometric Prior*

Table 3 compares the architectures under a common evaluation. The two models without a geometric prior reach 51.39 % and 48.92 % recovery at P90. The smallest graph model reaches 58.0 % with an order of magnitude fewer parameters than either of them.

Neither capacity nor a more elaborate aggregation moves that figure. Widening HexGNN from 38 k to 399 k parameters changes recovery by less than a point in either direction. The two architectures with a wider receptive field perform three points *worse*: Graphormer reaches 55.17 % with 4.3 M parameters and the graph U-Net 55.22 % with 332 k, against 58.02 % for the 38 k reference. PNA, which combines several aggregation statistics scaled by node degree, matches the reference at 58.45 % rather than exceeding it, using four times as many parameters. The problem behaves as a local one, and the isotropic mean over immediate neighbours is enough to solve it.

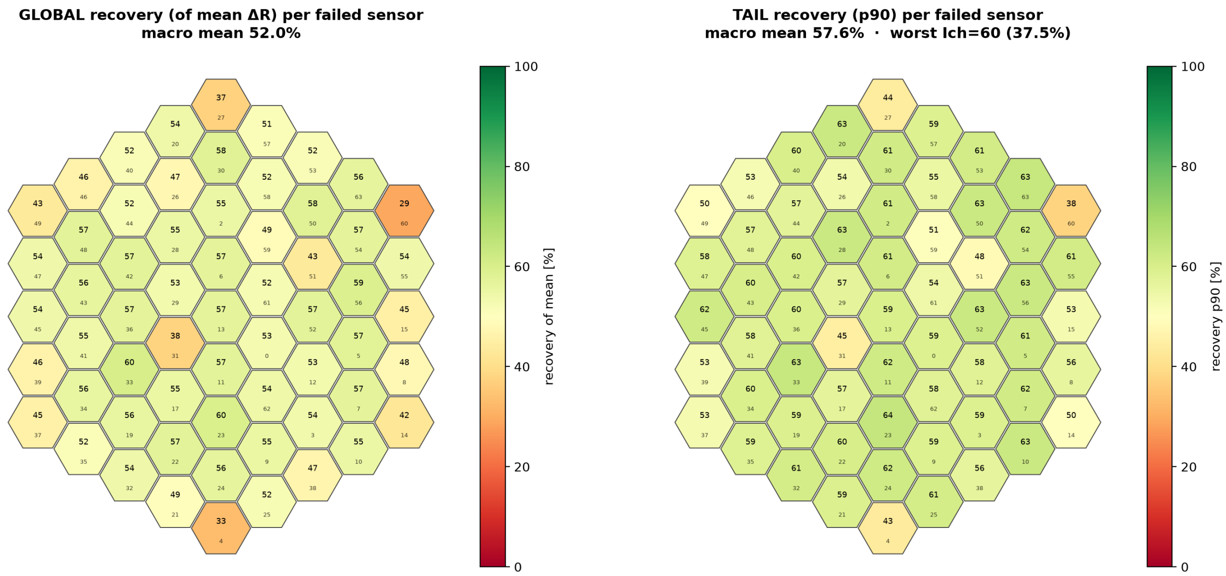

Figure 6: Recovery per channel over the 61 active sensors, at the 90th percentile (left) and on average (right). The colour scale is common to both panels. Recovery degrades towards the perimeter of the array.

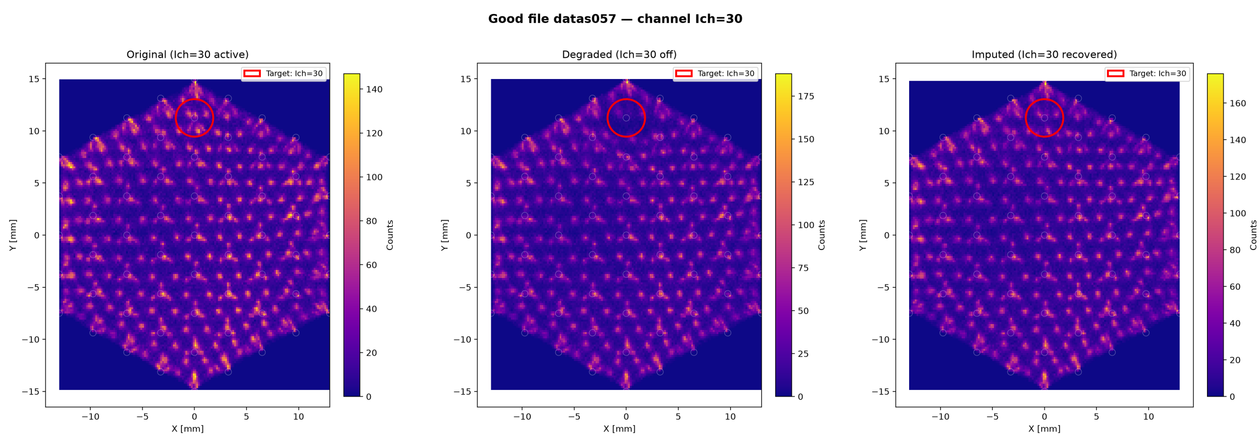

Figure 7: Flood map of a test module reconstructed from the same events. Left, with all 61 channels operational; centre, with one channel disabled; right, after imputing it.

Table 3: Architecture comparison under a common evaluation campaign. Recovery at P90 and on average, in per cent; MAE in normalised units.

Model	Params	Rec. p90	Rec. mean	MAE
DeepMLP	500 541	51.39	45.90	0.7219
ResidualMLP	345 981	48.92	44.81	0.7299
Graphormer	4 263 881	55.17	49.65	0.6600
Graph U-Net	331 649	55.22	47.95	0.6918
PNA	165 761	58.45	52.48	0.6309
HEXGNN-S	38 305	58.02	52.66	0.6195
HEXGNN-M	225 217	58.27	52.64	0.6185
HEXGNN-L	398 593	57.47	51.96	0.6385

To test whether the advantage comes from the detector geometry rather than from message passing in general, we retrained the reference with the adjacency randomly permuted, preserving the number of nodes, the number of edges and the degree distribution. The graph is then equally connected but physically meaningless. Recovery collapses to 28.08 %, 27.66 % and 31.28 % across three independent permutations, roughly half of the 58.0 % obtained with the true adjacency. The prior therefore accounts for about thirty points of the sixty the model achieves, and it is the specific adjacency of the detector that matters, not the fact of using a graph.

Against methods without learned parameters the network recovers 57.9 %, compared with 45.41 % for the method that applies the mean of the neighbouring channels and 46.56 % for a per-channel linear regression on the remaining 60. The margin is 12.49 and 11.33 percentage points respectively, which answers the question of whether the complexity of the model is warranted.

Table 4: Sources of variability in the reference model. Values are mean $\pm$ standard deviation over n independent runs.

Source of variation	n	**Rec. p90**	**Rec. mean**
Training run, fixed partition	4	58.04 ± 0.37	52.15 ± 0.31
Data partition	5	56.56 ± 2.32	50.60 ± 2.04

The linear baselines also let us measure how far the useful information extends. Restricting the regression to the first ring of neighbours, 5.1 channels on average, gives 44.04 % recovery. Adding the second ring, 13.7 channels, raises it to 46.42 %. The third ring adds 0.09 points and using all 60 remaining channels adds a further 0.04. Beyond the second ring there is essentially nothing left to extract, which is consistent with a model whose layers propagate information only a few hops.

3.4 Generalisation and Variability

To characterise the uncertainty of the reported figures we measured the two sources of variability separately (Table 4). Both refer to the reference configuration used throughout this section: HEXGNN-S trained with the MSE on single-channel failures, evaluated on the five held-out modules. Across four independent training runs on a fixed partition, differing in the global seed and in the order in which the modules are presented, recovery at P90 is 58.04 ± 0.37 %.

Repeating the experiment with five different partitions of the same 159 modules, however, gives 56.56 ± 2.32 %, a standard deviation six times larger. Evaluating the model on individual unseen modules explains why: recovery ranges from 29.2 % to 63.0 % depending on which detector it is measured on, with a standard deviation of 7.86 percentage points across modules. With five modules in the test partition, that places the standard error of the reported mean at 3.5 points.

Which of the two bands applies depends on the question. When two models are compared, the comparison is paired — the same modules, the same events and the same masks — so the band that applies is the ± 0.37 of the training run, and differences of tenths of a point are real. When the question is instead how the method will behave on a detector nobody has measured, the band that applies is the ± 2.32 of the partition, because what dominates there is which detectors happen to fall in the test set and not anything about the training. We therefore take 56.56 ± 2.32 % as the figure that describes what the method achieves on unseen hardware.

The minimum over the 61 channels behaves in the same way, only more so. Over four independent runs it averages 42.29 ± 4.45 %, spanning 37.54 % to 47.76 %, and the channel that attains it is not the same in every run: Ich 60 in two of them and Ich 31 in the other two. Its deviation across runs is twelve times that of the macro average computed over the same four runs.

3.5 Multiple Simultaneous Failures

Two questions arise once more than one channel fails at a time: whether a model trained on single failures still works, and whether the geometry used to generate the training masks matters. They are separate questions, and Table 5 answers both. Its rows are the geometry a model was *trained* on and its columns the geometry it is *evaluated* on, so the diagonal is each model on its own terrain.

Table 5: Recovery at P90 averaged over $k = 1 \dots 4$ suppressed channels, over a campaign of 20 mask seeds. Rows are the failure geometry used to generate the training masks; columns the geometry used at evaluation.

	Evaluated on		
Trained on	**Cluster**	**Near**	**Scatter**
Single channel (n=3)	33.74 ± 11.68	38.28 ± 9.23	47.69 ± 4.95
Cluster, k=$1-4$ (n=1)	58.46	52.62	52.44
Near, k=$1-4$ (n=3)	58.31 ± 0.51	54.61 ± 0.58	53.43 ± 0.67

The answer to the first question is that it does not transfer. The model trained on single failures drops to 33.74 % on contiguous ones, against the 58.31 % of a model trained with one to four channels: a gap of 24.6 points, the largest effect we measured. Its standard deviation across runs also rises to 11.68 points, because it still performs adequately at $k = 1$ and collapses at $k = 4$, so the average hides two very different behaviours. Exposing the model to multiple failures during training is therefore what determines whether it survives them, and it costs nothing, since the multi-failure models lose no accuracy at $k = 1$.

The answer to the second question is that the training geometry matters much less. Comparing the last two rows, the model trained on *near* exceeds the one trained on *cluster* by 1.99 points when both are evaluated on *near*, a difference of 2.9 standard deviations. On the other two geometries they are indistinguishable: -0.14 points on contiguous failures and $+0.99$ on scattered ones, the latter at 1.3 standard deviations. Training on training on *near* therefore buys a small advantage on its own terrain and loses nothing elsewhere, which is why we use it as the production configuration.

The evaluation geometries themselves behave in a way that deserves care, because the relative and the absolute readings point in opposite directions. Table 6 follows a single model, the one trained on *near*, as k grows. In recovery, contiguous failures are the *easiest* case for every $k \geq 2$: 57.7 % at $k = 4$ against 50.7 % for scattered ones. In millimetres they are the worst: four contiguous channels displace the centroid by 1.480 mm at P90, twice as much as four scattered ones, and after imputation still leave 0.628 mm against 0.366 mm.

Both readings come from the same runs, and the reason they disagree is that recovery is a ratio. A fault that causes more damage offers more damage to undo, so a larger repaired fraction can coexist with a larger residual error. Scattered failures are the least damaging in absolute terms and, at the same time, the hardest to correct proportionally. Fig. 8 shows both quantities as a function of k.

Table 6: Recovery and residual displacement as the number of suppressed channels grows, for the model trained on *near* ($n = 3$) evaluated under the three geometries. Displacements at P90, in millimetres.

	Recovery (%)			**Residual ΔR_{imp} (mm)**		
k	Clus.	Near	Scat.	Clus.	Near	Scat.
1	56.2	56.2	56.2	0.310	0.310	0.310
2	**59.5**	55.4	55.2	0.464	0.375	**0.305**
3	**59.9**	53.5	51.6	0.532	0.444	**0.366**
4	**57.7**	53.3	50.7	0.628	0.529	**0.366**

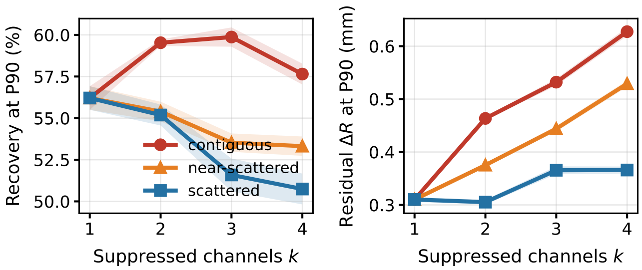

Figure 8: Recovery and residual displacement as the number of simultaneously suppressed channels grows, for the model trained on *near* and evaluated under the three failure geometries. Left, recovery at P90; right, the displacement that remains after imputation. Lines are the mean over three independently trained models and the shaded band their standard deviation. The two panels show the opposite readings discussed in the text: contiguous failures give the highest recovery and, at the same time, the largest residual error.

3.6 Physical Fidelity

Recovering the position is not the same as restoring a physically admissible signal, so we measured the two quantities the detector is built to deliver.

Separating the positioning error into its two components gives a result that a single aggregate figure would hide (Table 7). The systematic shift falls from 0.1262 mm to 0.0220 ± 0.0020 mm, so imputation removes 82.6 ± 1.6 % of it. The random spread, measured as the Gaussian-equivalent FWHM of Eq. 7, falls from 0.1656 mm to 0.0872±0.0015 mm, a reduction of only 47.3 ± 0.9 % (Fig. 9).

The asymmetry between the two is consistent with what the model can and cannot know. A systematic shift is predictable from the surviving channels, because it depends on which channel is missing and on where in the crystal the event occurred, both of which the network sees. The random component is the part of the lost charge that the neighbours do not determine, and no amount of information from them recovers it. After imputation the error distribution is therefore almost centred on the true position, and what remains is scatter around it.

The energy spectrum is preserved to a comparable degree, with a global spectral recovery of 84.7 ± 0.9 % ($n = 3$, Fig. 10). Here the core-to-edge gradient reappears: $87.4 \pm$

Table 7: Components of the positioning error before and after imputation. Mean $\pm$ standard deviation over three independently trained models; the degraded column does not depend on the model.

	Degraded (mm)	**Imputed (mm)**	**Removed (%)**
Systematic shift	0.1262	0.0220 ± 0.0020	82.6 ± 1.6
Random spread	0.1656	0.0872 ± 0.0015	47.3 ± 0.9

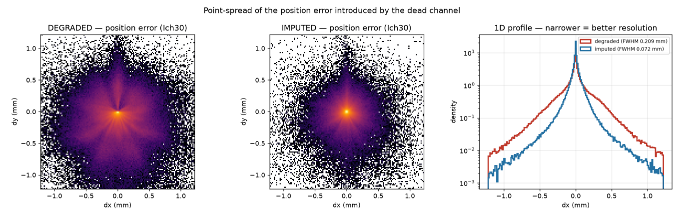

Figure 9: Distribution of the positioning error introduced by a suppressed channel, before and after imputation. There are two conditions and not three: the intact event is the reference against which the error is measured, so its error is zero by construction. The width of each distribution is the quantity reported as FWHM in Eq. 7.

1.8 % in the core, 85.6 ± 0.7 % in the middle band and 81.2 ± 0.3 % at the edge. The residual shift of the photopeak stays at 0.21 ± 0.02 ADC units across the three zones, against the 1.6 to 1.9 ADC units that the fault introduces before correction.

Both figures so far concern a single suppressed channel. Repeating the spectral measurement as k grows gives a result we did not expect (Table 8): the spectrum is *not* progressively lost. Recovery goes from 83.1 ± 0.2 % with one channel to 88.7 ± 0.3 % with four. The reason is again that recovery is a ratio. Suppressing four channels shifts the photopeak by 4.78 ADC units instead of 1.78, and the residual shift after imputation grows from 0.24 to 0.46, so the damage roughly triples while what remains of it only doubles.

The three failure geometries also reorder themselves with respect to the positioning results. At $k = 4$ the contiguous geometry gives the lowest spectral recovery, 85.6 ± 4.8 %, and the scattered one the highest, 93.3 ± 0.4 %, which is the opposite of the ordering in Table 6. In absolute terms, however, the two metrics agree: contiguous failures leave the largest residual in both, 0.628 mm of displacement and 0.628 ADC units of photopeak shift, and scattered ones the smallest. Compact groups remove a large amount of light from one region of the crystal, which is hard to restore in energy, while spreading the same four failures across the array keeps each individual deficit small. The *near* geometry sits between the two in both metrics and is the most stable of the three, with a standard deviation of 0.3 points against the 4.8 of the contiguous case.

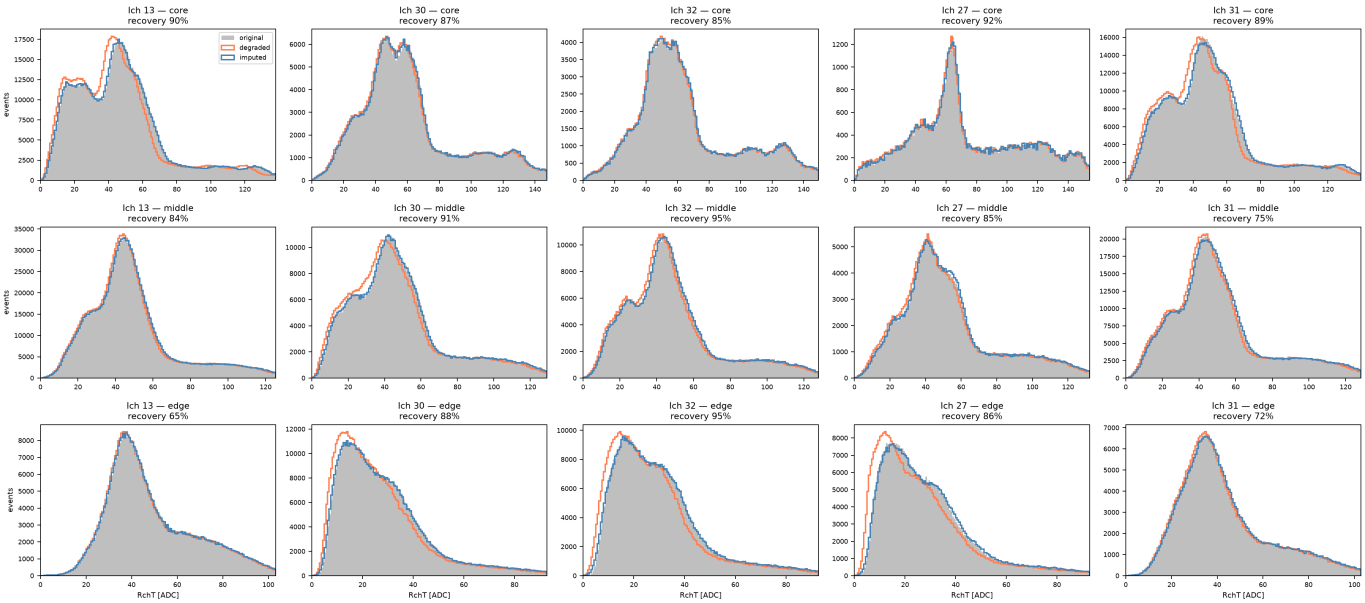

Figure 10: Energy spectra of the affected events for five channels (columns) and the three crystal zones (rows), comparing the original distribution in grey with the degraded one in orange and the imputed one in blue. The 511 keV photopeak of ^{22}Na is visible in the core and middle rows and flattens towards the edge, where the absorbing adhesive removes part of the collected light. The degraded curve departs from the original towards lower energies, and imputation brings it back; the per-panel figure is the spectral recovery.

Table 8: Spectral recovery as the number of suppressed channels grows, and by failure geometry at $k = 4$. Mean $\pm$ standard deviation over three models trained on *near*. Photopeak shifts in ADC units.

	Recovery (%)	**Shift deg.**	**Shift imp.**
$k = 1$	83.1 ± 0.2	1.78	0.24
$k = 2$	87.2 ± 0.8	2.80	0.34
$k = 3$	90.1 ± 1.7	3.88	0.33
$k = 4$	88.7 ± 0.3	4.78	0.46
$k = 4$, cluster	85.6 ± 4.8	5.03	0.63
$k = 4$, near	88.7 ± 0.3	4.78	0.46
$k = 4$, scatter	93.3 ± 0.4	3.62	0.20

3.7 Variants That Did Not Improve the Model

Four of the alternatives outlined in Sections 2.4 and 2.5 failed to improve on the reference. We report them with the form each one took, because in three of the four the reason for the failure lies in that form rather than in the idea behind it.

3.7.1 Physics-informed terms: Both variants add a penalty to the loss of Eq. 3,

$$\mathrm{Loss} = \mathrm{Loss}_{\mathrm{MSE}} + \lambda\,\Phi, \tag{8}$$

with Φ the physical quantity to preserve. We calibrated λ in each case so that both terms contribute comparably at the start of training.

The first uses the squared displacement of the centroid, $\Phi_{\Delta R} = \Delta R^2$ with $\lambda = 0.01$, computed with the same charge-weighted centroid as the evaluation metric. It changes recovery from 58.0 % to 58.2 %, inside the run-to-run band. The term turns out to be largely redundant: for a given channel, charge and position are related monotonically, so penalising the position error adds little that the MSE on the charge was not already penalising.

The second targets the energy spectrum, $\Phi_E = |\Delta R_T|$ with $\lambda = 0.7$, where ΔR_T is the shift in mean total charge. This one is actively harmful: recovery falls to 46.0 % and the bias changes sign. The cause is the form of the term and not the quantity it targets. The shift is computed over a batch, so the penalty acts on a *signed average* and the model can reduce it by letting errors of opposite sign cancel between events. It optimises the aggregate while degrading every individual event, which is exactly what the per-event metrics detect. A physics-informed term for this problem has to be computed per event.

3.7.2 Heteroscedastic prediction: Here the network outputs a mean and a log-variance per channel and trains on the Gaussian negative log-likelihood,

$$\mathrm{Loss}_{\mathrm{NLL}} = \frac{(y_c - \mu_c)^2}{2\sigma_c^2} + \log \sigma_c, \tag{9}$$

which would let the model flag the events it cannot impute reliably. It reaches 20.1 %. The likelihood offers two ways of lowering the loss, improving μ or inflating σ, and with nothing tying σ to a calibration target the second is cheaper. The model learns to declare uncertainty instead of learning to predict.

3.7.3 Ensembles: Averaging the outputs of several models is the only variant that exceeds the reference band, and it does so modestly. We tested two compositions. The first averages three replicas of the reference, identical except for the random seed, and reaches 58.69 %. The second averages three different architectures — HEXGNN with isotropic mean aggregation, PNA with multiple aggregators and a directional hexagonal convolution — on the assumption that their errors would be less correlated, and reaches 58.73 %. Against 58.04 ± 0.37 % for a single model, both gain about 0.7 points, and the heterogeneous composition adds almost nothing over the homogeneous one. The cost is three forward passes per event.

4 Discussion

4.1 What the Method Achieves

The question posed in Section 1 was whether the signal of a failed channel can be reconstructed instead of discarded. It can. On detectors the network has never seen, imputation removes 56.6 ± 2.3 % of the positioning error that the fault introduces, leaving a residual displacement of 0.28 mm at P90 — a 7.5 % of the distance between neighbouring sensors. The energy spectrum survives the operation at 84.7 ± 0.9 %, with a photopeak displaced by 0.21 ADC units where the uncorrected fault displaces it by 1.8.

Two properties of the approach matter as much as those numbers. The supervision is generated from healthy modules, so no detector with a genuinely dead sensor is needed to train, and the correction is applied before the centre of gravity is computed, so it returns a complete 61-channel vector that any downstream stage consumes unchanged.

4.2 Why a Small Network Suffices

The result that most constrains the interpretation is that a graph network of 38 305 parameters matches or exceeds every alternative we tried, including a graph transformer with a hundred times more parameters. This is not evidence that attention mechanisms are ineffective in general; it is evidence about the problem. The information that determines the charge of a missing channel is concentrated in its immediate surroundings, and once a model has access to it, additional capacity has nothing further to extract.

Three independent measurements support that reading. Restricting the linear baselines to concentric rings shows that the second ring already captures essentially everything: the third adds 0.09 points and the remaining 47 channels a further 0.04. Architectures with a wider receptive field perform three points worse rather than better, which is what one expects when the extra context is mostly noise. And permuting the adjacency while preserving the degree distribution collapses recovery from 58 to 28 %, so what the model exploits is the specific geometry of the detector and not the inductive bias of message passing.

This is consistent with the physics. Light sharing in a monolithic crystal is a local phenomenon: the charge collected by a sensor is determined by the interactions that occur near it, and the redundancy that makes imputation possible lives in that neighbourhood.

4.3 What Limits Performance

If capacity is not the constraint, the natural question is what is. Our measurements point away from the model and towards the data. Nine architectural variants, a doubled training budget and a larger parameter count all leave recovery within a point of the reference. What does move it, and by far, is which detector the model is measured on: recovery ranges from 29 to 63 % across individual modules, with a standard deviation of 7.9 points, against 0.3 between training runs on the same partition.

That spread is a property of the hardware, not of the method, and it is the price of working with real acquisitions. The 159 modules were built and assembled one by one, and they differ in ways the network never observes: the gain of each individual photosensor, the quality of the optical coupling between crystal and array, the exact placement of the source during each acquisition. No energy calibration was applied, so the charge the model reads is in ADC units whose scale is not common to all modules. A model trained across the ensemble can only learn what the ensemble shares; whatever is specific to one detector is, by construction, outside its reach.

There is a second consequence of using real data, and it bounds what can be verified rather than what can be learned. The fault we inject is synthetic because a real one leaves no reference: on a module whose channel is genuinely dead there is no way to know what that channel should have read, and therefore no way to measure the error of the imputation. Everything reported here is measured on healthy hardware with a fault we created, and the agreement between that setting and a real failure is an assumption, not a measurement. It is a reasonable assumption — the mask reproduces patterns we observed in the 62 defective modules — but it is worth stating as one.

Read together, the two points set the terms in which any figure in this work should be taken. Comparisons between models are precise, because they are paired over the same modules, events and masks. The absolute value on a new detector carries the variability of the hardware, and the way to reduce it is a per-module calibration rather than a better network.

4.4 Behaviour Under Multiple Failures

Training on single failures does not transfer to multiple ones, and the gap, 24.6 points, is the largest effect we measured. The geometry used to build the training masks matters far less: training on *near* gains 1.99 points on its own terrain and is indistinguishable from *cluster* elsewhere. The practical recommendation is therefore simpler than we expected — train with one to four channels suppressed and do not worry much about the pattern.

Two things about the energy did surprise us. The spectrum does not degrade as failures accumulate: recovery rises from 83 to 89 % between one and four suppressed channels, because the metric is a ratio and the damage grows faster than the residual. And the three geometries reorder themselves with respect to the positioning results, contiguous failures being the easiest case in position and the hardest in energy. A compact group displaces the centroid greatly but predictably, since the ring of surviving sensors around it still describes where the light went, while removing that much charge from one region is what makes the energy hard to restore. In absolute terms both metrics agree that a contiguous group is the worst case; it is only as a ratio that they disagree. The practical reading is that the energy information degrades gracefully, which is what a downstream energy window needs.

4.5 Relation to Existing Practice

Industrial quality control identifies dead pixels and compensates for them downstream, either by duplicating events from neighbouring pixels in list mode or by interpolating in sinogram space [22]. The channel itself is never recovered. The approach taken here intervenes earlier, before the centre of gravity is computed, and therefore preserves the crystal identification that downstream compensation cannot restore.

It also differs from the deep learning work closest to it. Networks trained to invert multiplexed readout schemes reconstruct individual channel values from a reduced set of measurements [27], [28], and the approach extends to semi-monolithic detectors where the full response is recovered from row and column sums [29]. In all of those cases the information is present in the acquired data, encoded in a linear combination fixed by design, and the network learns to invert a known mapping. A failed channel is a different problem: nothing of its signal reaches the acquisition, and the only remaining trace is the statistical correlation that light sharing induces among its neighbours.

That last point is what connects this work with the other deep learning studies on this same detector. Both estimate the interaction position from the 61 channels, one on simulated data [18] and one on measured [26], [44]. Ours does not compete with them: it repairs the 61-channel vector that they take as input, and the two are consecutive stages of the same chain. What they have in common is more interesting than what separates them. All three exploit the fact that in a monolithic crystal the light of one event reaches many sensors, so the array carries more information than the number of channels suggests — position between sensors in their case, the value of a missing sensor in ours. A pixelated detector offers neither: its segmentation fixes the spatial resolution at the pixel pitch, and a dead pixel is a hole with no redundancy to fill it from. That redundancy is difficult to exploit analytically, which is historically why monolithic designs were set aside, and it is exactly what a learned model is good at. The results reported here suggest that this is also where their remaining margin lies.

4.6 Limitations

Three limitations bound what the preceding results support, beyond the two properties of the data discussed in Section 4.

The test partition contains five modules. Given the variability between detectors, that places the standard error of any absolute figure at approximately 3.5 points, which is why we report cross-validated values rather than the result of a single partition.

The simulated fault is a total suppression. Real degradation is often partial: a channel that keeps part of its response, or that drifts with temperature. The method has not been evaluated in that regime, and it is not obvious that a model trained on complete failures transfers to it.

Finally, the minimum over the 61 channels proved unusable as a robustness metric, and the reason is instructive rather than mysterious. Its variation across training runs, ± 4.5 points, exceeds any difference we could have attributed to a design choice, and the channel attaining it changes from run to run. A minimum over 61 values is a noisy statistic to begin with, and here it lands on whichever channel happens to be worst served by the data: the sensors do not all have the same gain, the source was not equidistant from all of them, and some channels simply receive less usable light than others in a given acquisition. What the metric measures is therefore closer to the luck of the acquisition than to the robustness of the model. Separating the two would require data in which the per-channel response is known or controlled, which real flood acquisitions do not provide.

4.7 Future Work

Four directions follow from the above, and the first two address the two limits identified in Section 4 rather than any shortcoming of the network.

A calibrated acquisition would separate what belongs to the method from what belongs to the hardware. With per-channel gains known, the charge of every module would live on a common scale, the spectral metric could be reported in keV instead of as a conservation ratio, and the spread between detectors could be attributed rather than merely measured. A complementary route is a controlled simulation of this same geometry, where the response of each sensor is set rather than inferred: it would make the worst-channel behaviour analysable, and it would allow the synthetic fault to be validated against a known reference.

The second is the 62 modules with genuine defects, of which 18 are annotated so far. They cannot give a quantitative error, for the reason given above, but they can answer a question the synthetic experiments cannot: whether an imputation trained on injected faults produces flood maps that an expert accepts as repaired.

The third is partial degradation. Extending the training masks to channels that keep part of their response, rather than none of it, would cover the failure mode that the current model does not see and that the annotation campaign suggests is the more common one in practice.

The fourth concerns what the imputation is for. Everything here is measured on single detectors, and the quantity that matters clinically is the resolution of the reconstructed image. Propagating imputed events through a full reconstruction, and quantifying what a repaired module contributes to a ring of them, is the step that would turn a per-detector correction into a system-level result.

5 Conclusions

We have presented a deep learning method that recovers the signal lost to a faulty SiPM channel in a monolithic PET detector instead of discarding it. Monolithic scintillators are cheaper to produce than pixelated arrays and they do not impose a segmentation on the information that reaches the reconstruction stage, but they pay for it with a readout in which every sensor contributes to every event: when one channel fails, the flood map degrades across the whole crystal and the module is normally set aside. Recovering that channel is therefore a way of keeping detectors, and the events they have already acquired, that current practice gives up.

The method is a graph neural network, HexGNN, built on the geometry of the array rather than on top of it: one node per sensor, one edge between adjacent sensors, and message passing along the same paths through which the light is shared. It has 38 305 parameters. Its supervision comes from healthy modules — we suppress a channel whose true value we know — so no detector with a genuine fault is needed in order to train it, and the correction is applied before the centre of gravity is computed, which returns a complete 61-channel vector that the rest of the processing chain consumes unchanged.

On modules the network had never seen, imputation removes $56.6 \pm 2.3\,\%$ of the positioning error introduced by the fault, leaving a residual displacement of 0.28 mm at P90, and it preserves the energy spectrum to $84.7 \pm 0.9\,\%$. Training with one to four simultaneous failures, following the patterns we found in the flood maps of the modules that arrived with a defect, extends that behaviour to several dead channels at no cost in the single-channel case.

Three lines follow most directly from this work. A calibrated acquisition would separate what belongs to the method from what belongs to the hardware. The modules that arrived with a defect offer a validation against expert criterion that no injected fault can give. And propagating imputed events through a full reconstruction would take the correction from the detector to the system. What the work establishes is that the information needed to repair a dead channel is there in its neighbours, and that a small network with the geometry of the detector built into it is enough to reach it.

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